CS409 Final term Handouts

Topic 86

What is Free Space in Data Block?

- As the database fills a data block from the bottom up, the amount of free space between the row data and the block header decreases. This free space can also shrink during updates, as when changing a trailing null to a non-null value.
- The database manages free space in the data block to optimize performance and avoid wasted space (Tuning Feature).

Data Blocks & OS Blocks

- At the physical level, database data is stored in disk files made up of operating system blocks.
- An operating system block is the minimum unit of data that the operating system can read or write.
- In contrast, an Oracle block is a logical storage structure whose size and structure are not known to the operating system.

Percentage of Free Space in Data Blocks

- The PCTFREE storage parameter is essential to how the database manages free space. This SQL parameter sets the minimum percentage of a data block reserved as free space for updates to existing rows.
- PCTFREE is important for preventing row migration and avoiding wasted space.

Optimizing of Free Space

- While the percentage of free space cannot be less than PCTFREE, the amount of free space can be greater. For example, a PCTFREE setting of 20% prevents the total amount of free space from dropping to 5% of the block, but permits 50% of the block to be free space.

The following SQL statements can increase free space:

- DELETE statements
- UPDATE statements that either update existing values to smaller values or increase existing values and force a row to migrate

- INSERT statements on a table that uses OLTP compression. If inserts fill a block with data, then the database invokes block compression, which may result in the block having more free space.
- The space released is available for INSERT under the following conditions:
- If the INSERT statement is in the same transaction and after the statement that frees space, then the statement can use the space.
- If the INSERT statement is in a separate transaction from the statement that frees space (perhaps run by another user), then the statement can use the space made available only after the other transaction commits and only if the space is needed.

Reuse of Index Space

- The database can reuse space within an index block.
- For example, if you insert a value into a column and delete it, and if an index exists on this column, then the database can reuse the index slot when a row requires it

Topic 87

What is Extent?

An extent is a logical unit of database storage space allocation made up of contiguous data blocks. Data blocks in an extent are logically contiguous but can be physically spread out on disk because of RAID striping and file system implementations.

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Deallocation of Extents

In general, the extents of a user segment do not return to the tablespace unless you

drop the object using a DROP command. In Oracle Database 11g Release 2, you can also drop the segment using the DBMS_SPACE_ADMIN package.

For example, if you delete all rows in a table, then the database does not reclaim the data blocks for use by other objects in the tablespace.

Manually Deallocate Space

Following techniques can deallocate extents

- You can use an online segment shrink to reclaim fragmented space in a segment. Segment shrink is an online, in-place operation.
- You can move the data of a non-partitioned table or table partition into a new segment, and optionally into a different tablespace for which you have quota.
- You can rebuild or coalesce the index
- You can truncate a table or table cluster, which removes all rows. By default, Oracle Database deallocates all space used by the removed rows except that specified by the MINEXTENTS storage parameter.
- You can deallocate unused space, which frees the unused space at the high water mark end of the database segment and makes the space available for other segments in the tablespace

Sample SQL DDL using Extent

```
CREATE TABLESPACE MCare_indx

DATAFILE 'D:\oracle11g\ora11g\oradata\MCare_indx01.dbf'

SIZE 100M

MINIMUM EXTENT 64K

DEFAULT STORAGE (INITIAL 64K NEXT 64K);
```

Topic 88

What is Segment?

A segment is a set of extents that contains all the data for a logical storage structure within a tablespace. For example, Oracle Database allocates one or more extents to form the data segment for a table. The database also allocates one or more extents to form the index segment for a table implementation.

User Segments

A single data segment in a database stores the data for one user object. There are different types of segments. Examples of user segments include:

- Table, table partition, or table cluster
- LOB or LOB partition
- Index or index partition

Each non partitioned object and object partition is stored in its own segment. For example, if an index has five partitions, then five segments contain the index data.

User Segments Creation

By default, the database uses deferred segment creation to update only database metadata when creating tables and indexes.

When a user inserts the first row into a table or partition, the database creates segments for the table or partition, its LOB columns, and its indexes.

Deferred segment creation enables you to avoid using database resources unnecessarily.

For example, installation of an application can create thousands of objects, on consuming significant disk space. Many of these objects may never be used.

You can use the DBMS_SPACE_ADMIN package to manage segments for empty objects.

SQL DDL for Segments under Tablespace

CREATE TABLE PATIENT_HISTORY

(PATIENT_ID CHAR (007),

DEPEND_SNO VARCHAR2 (002),

ALLERGY_DESC VARCHAR2 (250) NOT NULL,

HISTORY_DESC LONG

) tablespace MCare;

ALTER TABLE PATIENT_HISTORY

ADD constraint PK_PATHIST_ID PRIMARY KEY (PATIENT_ID,DEPEND_SNO)

USING INDEX TABLESPACE MCare_indx;

Temporary Segments

When processing a query, Oracle Database often requires temporary workspace for

intermediate stages of SQL statement execution.

Typical operations that may require a temporary segment include sorting, hashing, and merging bitmaps. While creating an index, Oracle Database also places index segments into temporary segments and then converts them into permanent segments when the index is complete

UNDO Segments

Oracle Database maintains records of the actions of transactions, collectively known as undo data.

Oracle Database uses undo to do the following:

- Roll back an active transaction
- Recover a terminated transaction
- Provide read consistency
- Perform some logical flashback operations

Tablespace with Segments

A tablespace is a logical storage container for segments. Segments are database objects, such as tables and indexes, that consume storage space. At the physical level, a tablespace stores data in one Or more data files or temp files.

Topic 89

Introduction to Schema Objects

A database schema is a logical container for data structures, called schema objects.

Examples of schema objects are tables, views, sequences and indexes etc.

Schema objects are created and manipulated with SQL

A database user has a password and various database privileges. Each user owns a single schema, which has the same name as the user.

The schema contains the data for the user owning the schema. For example, the hr user owns the hr schema, which contains schema objects such as the employees table.

In a production database, the schema owner usually represents a database application rather than a person.

Schema Objects Types

- Tables

  Storing data in rows of tables

- Indexes

  Indexes are schema objects that contains an entry for each indexed row of the table

  or table cluster and provide direct, fast access to rows.

- Partitions

  Partitions are pieces of large tables and indexes.

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- Views

  Views are customized presentations of data in one or more tables or other views.

- Sequences

  A sequence is a user-created object that can be shared by multiple users to generate integers.

- Dimensions

  A dimension defines a parent-child relationship between pairs of column sets,

  where all the columns of a column set must come from the same table.

- Synonyms

  A synonym is an alias for another schema object.

- PL/SQL Programs

Procedural server programming

Overview of Views

- A view is a logical representation of one or more tables.
- In essence, a view is a stored query.
- A view derives its data from the tables on which it is based, called base tables.
- Base tables can be tables or other views.
- All operations performed on a view actually affect the base tables.
- You can use views in most places where tables are used.

Use of Views

- Provide an additional level of table security by restricting access to a predetermined set of rows or columns of a table
- Hide data complexity
- Present the data in a different perspective from that of the base table

For example, the columns of a view can

be renamed without affecting the

tables on which the view is based.

- Isolate applications from changes in definitions of base tables used.

SQL to Create View

- CREATE VIEW staff AS
- SELECT employee_id, last_name, job_id, manager_id, department_id
- FROM employees;

Topic 90

<u>Reading Material:</u>

Overview of Table Clusters

A table cluster is a group of tables that share common columns and store related data in the same blocks. When tables are clustered, a single data block can contain rows from multiple tables.

For example, a block can store rows from both the employees and Departments tables rather than from only a single table.

Cluster Key

- The cluster key is the column or columns that the clustered tables have in common.
- For example, the employees and departments tables share the department_id column.
- You specify the cluster key when creating the table cluster and when creating every table added to the table cluster.

Cluster Key Value

- The cluster key value is the value of the cluster key columns for a particular set of rows. All data that contains the same cluster key value, such as department_id=20, is physically stored together. Each cluster key value is stored only once in the cluster and the cluster index, no matter how many rows of different tables contain the value

Overview of Indexed Clusters

An indexed cluster is a table cluster that uses an index to locate data. The cluster index is a B-tree index on the cluster key. A cluster index must be created before any rows can be inserted into clustered tables

DDL for Indexed Clusters

CREATE CLUSTER emp_dept_cluster

(department_id NUMBER(4))

SIZE 512;

CREATE INDEX idx_emp_dept_cluster ON CLUSTER emp_dept_cluster;

Overview of Indexes

An index is an optional structure, associated with a table or table cluster, that can sometimes speed data access. By creating an index on one or more columns of a table, you gain the ability in some cases to retrieve a small set of randomly distributed rows from the table.

Indexes are one of many means of reducing disk I/O.

Composite Indexes

A composite index, also called a concatenated index, is an index on multiple columns in a table. Columns in a composite index should appear in the order that makes the most sense for the queries that will retrieve data and need not be adjacent in the table.

CREATE INDEX employees_ix

ON employees (last_name, job_id, salary);

Topic 91

Reading Material:

Data Integrity

Business rules specify conditions and relationships that must always be true or must always be false. For example, each company defines its own policies about salaries, employee numbers, inventory tracking, and so on. It is important that data maintain **data integrity**, which is adherence to these rules, as determined by the database administrator or application developer.

Techniques for Data Integrity

Following are possible options:

- Enforcing business rules with triggered stored database procedures
- Using stored procedures to completely control access to data
- Enforcing business rules in the code of a database application
- Using Oracle Database integrity constraints, which are rules defined at the column or object level that restrict values in the database

Advantages of Integrity Constraints

An integrity constraint is a schema object that is created and dropped using SQL. To enforce data integrity, use integrity constraints unless it is not possible. Advantages of integrity constraints over alternatives for enforcing data integrity include:

■Declarative ease

Because you define integrity constraints using SQL statements, no additional programming is required when you define or alter a table. The SQL statements are easy to write and eliminate programming errors.

■Centralized rules

Integrity constraints are defined for tables and are stored in the data dictionary. Thus, data entered by all applications must adhere to the same integrity constraints. If the rules change at the table level, then applications need not change. Also, applications can use metadata in the data dictionary to immediately inform users of violations, even before the database checks the SQL statement.

■Flexibility when loading data

You can disable integrity constraints temporarily to avoid performance overhead when loading large amounts of data. When the data load is complete, you can re-enable the integrity constraints.

Types of Integrity Constraints

Oracle Database enables you to apply constraints both at the table and column level. A constraint specified as part of the definition of a column or attribute is called an inline specification.

NOT NULL

UNIQUE

CHECK

DEFAULT

PK/ FK: Standard Integrity Constraints

Topic 92

<u>Reading Material:</u>

Managing Integrity Constraints-2

NOT NULL Constraint

- A NOT NULL constraint requires that a column of a table contain no null values. A null is the absence of a value. By default, all columns in a table allow nulls.
- By default, FK is NULL.

Create table Student(

(RollNo CHAR(10),

FullName VARCHAR2(50) NOT NULL,

…..);

UNIQUE Constraint

- A unique key (Null able) constraint requires that every value in a column or set of columns be unique. No rows of a table may have duplicate values in a column (the unique key) or set of columns (the composite unique key) with a unique key constraint.

Create table Medicine(

(ItemNo CHAR(10),

Name VARCHAR2(50) NOT NULL UNIQUE,

…..);

CHECK Constraint

- A check constraint on a column or set of columns requires that a specified condition be true or unknown for every row. If DML results in the condition of the constraint evaluating to false, then the SQL statement is rolled back.

Alter table Medicine

Add constraint chk_medType

CHECK (MedType IN ('S','P'));

DEFAULT Constraint

- A default constraint on a column requires that a specified value be true when null value is entered.

Create table Medicine(

(ItemNo CHAR(10),

Name VARCHAR2(50) NOT NULL,

Status CHAR(1) default 'Y‘,

…..);

FK CASCADE Constraints

- CASCADE is applied with INSERT, DELETE or UPDATE.
- For example, a foreign key with cascade delete means that if a record in the parent table is deleted, then the corresponding records in the child table will automatically be deleted. This is called a cascade delete in Oracle.

FK CASCADE Constraints - Example

```
CREATE TABLE supplier

( supplier_id numeric(10) not null,

  supplier_name varchar2(50) not null,

  contact_name varchar2(50),

  CONSTRAINT supplier_pk PRIMARY KEY (supplier_id));

CREATE TABLE products

( product_id numeric(10) not null,

  supplier_id numeric(10) not null,

  CONSTRAINT fk_supplier FOREIGN KEY (supplier_id)

    REFERENCES supplier(supplier_id)  ON DELETE CASCADE);
```

Trigger's Constraints

```
CREATE TRIGGER cascade_del_stu_record

BEFORE DELETE ON STUDENT

FOR EACH ROW

BEGIN

   DELETE FROM RECORD

   WHERE RECORD.SID = :OLD.SID;

END;

CREATE TRIGGER cascade_del_stu_record

BEFORE DELETE ON STUDENT
```

```
FOR EACH ROW

BEGIN

  DELETE FROM RECORD

  WHERE RECORD.SID = :OLD.SID;

END;
```

ENABLE/ DISABLE or Drop Constraints

Existing constraints on table can be

- Enabled
- Disabled
- Dropped

For example,

```
Alter table Invoice

DROP constraint fk_custid;
```

Topic 93

Reading Material:

How a Schema is Created

- We have studied that DBA has to create default Tablespace and assign temporary Tablespace with storage in terms of data files.
- Create first user (owner of schema) with grants such as rights and privileges.

How a Schema is Implemented?

- A view is a logical representation of a table or combination of tables.
- In essence, a view is a stored query.
- A view derives its data from the tables on which it is based.
- These tables are called base tables. Base tables might in turn be actual tables or might be views themselves.
- All operations performed on a view actually affect the base table of the view.

- You can use views in almost the same way as tables. You can query, update, insert into, and delete from views, just as you can standard tables.
- Views can provide a different representation (such as subsets or supersets) of the data that resides within other tables and views.
- Views can be Simple or Complex
- Views can be used for security purposes
- View does not contain its own data

Simple & Complex Views

Here is the example of sales staff view that have id 10

CREATE VIEW sales_staff AS

SELECT empno, ename, deptno

FROM emp

WHERE deptno = 10;

CREATE VIEW sales_staff AS

SELECT job, count(*)

FROM emp

WHERE deptno = 10

Group by job;

CREATE OR REPLACE VIEW division1_staff AS

SELECT ename, empno, job, dname

FROM emp, dept

WHERE emp.deptno IN (10, 30)

AND emp.deptno = dept.deptno;

Insert, delete and update operations for base tables are applied on simple views.

Creating Tables

CREATE TABLE dept (

deptno NUMBER(4) PRIMARY KEY,

dname VARCHAR2(14),

loc VARCHAR2(13));

CREATE TABLE emp (

empno NUMBER(4) PRIMARY KEY,

ename VARCHAR2(10),

…

deptno NUMBER(2);

…

Creating Sequence or AutoNumber

CREATE SEQUENCE emp_sequence

INCREMENT BY 1

START WITH 1

NOMAXVALUE

NOCYCLE

CACHE 10;

Data Dictionary for Tables & Views

SELECT COLUMN_NAME, UPDATABLE

FROM USER_UPDATABLE_COLUMNS

WHERE TABLE_NAME = 'EMP_DEPT';

Topic 94

Reading Material:

Creating Tables & Alter Tables

What is a Table?

- Tables are the basic unit of data storage in an Oracle Database. Data is stored in rows and columns. You define a table with a table name, such as employees, and a set of columns.
- You give each column a column name, such as emp_id, last_name, and job_id; a data type, such as VARCHAR2, DATE, or NUMBER; and a width. The width can be predetermined by the data type, as in DATE.
- If columns are of the NUMBER data type, define precision and scale instead of width. A row is a collection of column information corresponding to a single record.
- You can specify rules for each column of a table. These rules are called integrity constraints. One example is a NOT NULL integrity constraint. This constraint forces the column to contain a value in every row.
- Some column types, such as LOBs, varrays, and nested tables, are stored in their own segments. LOBs and varrays are stored in LOB segments, while nested tables are stored in storage tables.

Design Table Guidelines

- Use descriptive names for tables, columns, indexes, and clusters.
- Be consistent in abbreviations and in the use of singular and plural forms of table names and columns.
- Document the meaning of each table and its columns with the COMMENT command.
- Normalize each table.
- Select the appropriate datatype for each column. Consider whether your applications would benefit from adding one or more virtual columns to some tables.
- Define columns that allow nulls last, to conserve storage space.
- Cluster tables whenever appropriate, to conserve storage space and optimize performance of SQL statements.

Heap-Organized Table

- This is the basic, general purpose type of table which is the primary subject of this chapter. Its data (heap).

A heap-organized table is a table with rows stored in no particular order. This is a standard Oracle table; the term "heap" is used to differentiate it from an index-organized table or external table. If a row is moved within a heap-organized table, the row's ROWID will also change.

Clustered Table

A clustered table is a table that is part of a cluster. A cluster is a group of tables that share the same data blocks because they share common columns and are often used together.

Indexed-Organized Table

Unlike an ordinary (heap-organized) table, data for an index-organized table is stored in a B-tree index structure in a primary key sorted manner.

Besides storing the primary key column values of an index-organized table row, each index entry in the B-tree stores the non key column values as well.

Partitioned Table

Partitioned tables enable your data to be broken down into smaller, more manageable pieces called partitions, or even subpartitions. Each partition can have separate physical attributes, such as compression enabled or disabled, type of compression, physical storage settings, and tablespace, thus providing a structure that can be better tuned for availability and performance.

In addition, each partition can be managed individually, which can simplify and reduce the time required for backup and administration.

Storage Location of a Table

- It is advisable to specify the TABLESPACE clause in a CREATE TABLE statement to identify the tablespace that is to store the new table.
- For partitioned tables, you can optionally identify the tablespace that is to store each partition. Ensure that you have the appropriate privileges and quota on any tablespaces that you use.
- If you do not specify a tablespace in a CREATE TABLE statement, the table is created in your default tablespace.

Alter Table

- Alter table command applies when table is already exist in a schema.
- We can add constraints, columns and modifications operations on table using Alter table DDL statements.

Tables with Segment Creation

Lesson 95 : DDL Script Writing-1

Reading Material:

What is a Script?

Normally writing or generating DDL commands to create/ alter tables, views, constraints and storages etc., in an order in plain text using Notepad or text files

Mostly scripts contain structure of tables. Sometimes, DML commands are also put in scripts as per management requirement.

Scripts can be with Create, Alter and Drop Tables

Sample of Script:

Creating Temporary Tables

Temporary tables are useful in applications where a result set is to be buffered

(temporarily persisted), perhaps because it is constructed by running multiple DML

operations.

For example, consider the following:

Table with TEMP Tablespace

Bulk Data Load in a Table

Alter Table

ALTER TABLE statement is used to add a column, modify a column, drop a column, rename a column or rename a table.

Index with Tablespace

CREATE INDEX int_sales_index ON int_salestable

(s_saledate, s_productid, s_custid)

TABLESPACE tbs_low_freq;

Lesson 97 : Drop Tables & Recycle Bin

Reading Material:

Drop Tables

To drop a table that you no longer need, use the DROP TABLE statement. The table must be contained in your schema or you must have the DROP ANY TABLE system privilege.

Drop Tables Precautions

- Dropping a table removes the table definition from the data dictionary. All rows of the table are no longer accessible.
- All indexes and triggers associated with a table are dropped.
- All views and PL/SQL program units dependent on a dropped table remain, yet become invalid (not usable).
- All synonyms for a dropped table remain, but return an error when used.
- All extents allocated for a table that is dropped are returned to the free space of the tablespace and can be used by any other object requiring new extents or new objects. All rows corresponding to a clustered table are deleted from the blocks of the cluster

Drop Tables Commands

DROP TABLE hr.int_admin_emp;

- If the table to be dropped contains any primary or unique keys referenced by foreign keys of other tables and you intend to drop the FOREIGN KEY constraints of the child tables, then include the CASCADE clause in the DROP TABLE statement

DROP TABLE hr.admin_emp CASCADE CONSTRAINTS;

If you should want to immediately release the space associated with the table at the time you issue the DROP TABLE statement, include the PURGE clause as shown in the following statement:

DROP TABLE hr.admin_emp PURGE;

What is the Recycle Bin?

- The recycle bin is actually a data dictionary table containing information about dropped objects.
- Dropped tables and any associated objects such as indexes, constraints, nested tables, and the likes are not removed and still occupy space.
- They continue to count against user space quotas, until specifically purged from the recycle bin.
- SELECT * FROM RECYCLEBIN;

Object Naming in Recycle Bin

- The renaming convention is as follows:

  BIN$unique_id$version

- unique_id is a 26-character globally unique identifier for this object, which makes the recycle bin name unique across all databases
- Version is a version number assigned by the database

Enable & Disable the Recycle Bin

Purging is the Recycle Bin

If you decide that you are never going to restore an item from the recycle bin, you can use the PURGE statement to remove the items and their associated objects from the recycle bin and release their storage space.

PURGE TABLE "BIN$jsleilx392mk2=293$0";

PURGE TABLESPACE example;

PURGE RECYCLEBIN;

Data Dictionary Views for Recycle Bin

Lesson 98 : Indexed Organized Tables

Reading Material:

What is an index?

Indexes are optional structures associated with table and clusters that allow SQL queries to execute more quickly against a table

Index helps you locate information faster than if there were no index, an oracle database index provides a faster access path to table data. You can use indexes without rewriting any queries. Your results are the same, but you see them more quickly.

Independency of an index

Indexes are logically and physically independent of the data in the associated table. Being independent structure, they require storage space.

You can create and drop an index without affecting the base tables, database application or other indexes.

The database automatically maintains indexes when you insert, update and delete rows of the associated table.

If you drop an index all applications continue to work, access to previously indexed data might be slower.

Data load & index usage

Data is often inserted or loaded into a table using either a SQL * loader or an import utility. It is more efficient to create an index for a table after inserting or loading the data.

If you create one or more indexes before loading data, the database then must update every index as each row is inserted.

Index with segments & Tablespace

• The amount for each user is determined by the initialization parameter SORT_AREA_SIZE.

• The database also swaps sort information to and from temporary segments that are only allocated during the index creation in the user's temporary tablespace.

Suitable Columns for Indexing

• Values are relatively unique in the column.

• There is a wide range of values (good for regular indexes).

• There is a small range of values (good for bitmap indexes).

• The column contains many nulls, but queries often select all rows having a value.

Indexing Limit on Table

• A table can have any number of indexes. However, the more indexes there are, the more overhead is incurred as the table is modified Specifically, when rows are inserted or deleted, all indexes on the table must be updated as well.

• Also, when a column is updated, all indexes that contain the column must be updated.

Creating Indexes

To create an index in your own schema, at least one of the following conditions must be true:

• The table or cluster to be indexed is in your own schema.

• You have INDEX privilege on the table to be indexed.

• You have CREATE ANY INDEX system privilege.

CREATE INDEX emp_ename ON emp(ename);

CREATE INDEX emp_ename ON emp(ename) TABLESPACE users STORAGE (INITIAL 20K NEXT 2010;

CREATE UNIQUE INDEX dept_unique_index ON dept (dname) TABLESPACE indx;

Creating & Alter Indexes

CREATE TABLE emp (empno NUMBER(5) PRIMARY KEY, age INTEGER) ENABLE PRIMARY KEY USING INDEX TABLESPACE Work_Indx;

ALTER INDEX emp_ename STORAGE (NEXT 40);

Lesson 99 : Data Dictionary Views for Tables

<u>Reading Material:</u>

Data Dictionary Views for Tables

Three Types of Views

There are three types of Views to manage Tables related information:

DBA_

ALL_

USER_

Displaying Views for Tables

SQL statement to display Tables related Views:

Select Table_name

From Dict

Where Table_name like '%TAB%‘;

You can use following choices for more views:

Table_name like '%COL%'

Table_name like '%EXTERNAL%‘

Table_name like '%CONSTRAINT%'

Size of a Table

- To find size of table in MB

select sum(bytes)/1024/1024 SizeMB

from dba_segments

where segment_name ='EMP';

Displaying Table's Constraints

Displaying Table's Columns Details

Column information, such as name, data type, length, precision, scale, and default data values can be listed using one of the views ending with the _COLUMNS suffix. For example, the following query lists all of the default column values for the emp and dept tables:

SELECT TABLE_NAME, COLUMN_NAME, DATA_TYPE, DATA_LENGTH, LAST_ANALYZED

FROM DBA_TAB_COLUMNS

WHERE OWNER = 'HR'

ORDER BY TABLE_NAME;

Output of the above code:

Lesson 100 : Creating & Alter an Index

Reading Material:

Altering Indexes

To alter an index, your schema must contain the index or you must have the ALTER ANY INDEX system privilege. With the ALTER INDEX statement, you can:

- Rebuild or coalesce an existing index
- Deallocate unused space or allocate a new extent Specify parallel execution (or not) and alter the degree of parallelism
- Alter storage parameters or physical attributes Specify LOGGING or NOLOGGING

Creating Indexes

- Enable or disable key compression
- Mark the index unusable
- Make the index invisible
- Rename the index
- Start or stop the monitoring of index usage

CREATE INDEX emp_ename ON emp(ename)

TABLESPACE users

STORAGE (INITIAL 20K NEXT 20k);

CREATE UNIQUE INDEX dept_unique_index ON dept (dname) TABLESPACE indx;

Creating & Alter Indexes

CREATE TABLE emp (
empno NUMBER(5) PRIMARY KEY,

age   INTEGER)

ENABLE PRIMARY KEY

USING INDEX TABLESPACE Work_Indx;

ALTER INDEX emp_ename

STORAGE (NEXT 40);

Rebuild Index

You have the option of rebuilding the index online. Rebuilding online enables you to update base tables at the same time that you are rebuilding.

The following statement rebuilds the emp_name index online:

ALTER INDEX emp_name REBUILD ONLINE;

Making Index Visible/ Invisible

An invisible index is ignored by the optimizer unless you explicitly set the OPTIMIZER_USE_INVISIBLE_INDEXES initialization parameter to TRUE at the session or system level. Making an index invisible is an alternative to making it unusable or dropping it.

ALTER INDEX *index* INVISIBLE;

ALTER INDEX *index* VISIBLE;

Making Index Useable

Query the data dictionary to determine whether an existing index or index partition is usable or unusable.

SELECT INDEX_NAME AS "INDEX OR PART NAME", STATUS, SEGMENT_CREATED

FROM USER_INDEXES

UNION ALL

SELECT PARTITION_NAME AS "INDEX OR PART NAME", STATUS, SEGMENT_CREATED

FROM USER_IND_PARTITIONS;

Lesson 101 : Creating Hash Index

Reading Material:

What is the Recycle Bin?

The recycle bin is a data dictionary table containing information about dropped objects. Dropped tables and any associated objects such as indexes, constraints, nested tables, and the likes are not removed and still occupy space. They continue to count against user space quotas, until specifically purged from the recycle bin or the unlikely situation where they must be purged by the database because of tablespace space constraints.

Each user can be thought of as having his own recycle bin, because, unless a user has the SYSDBA privilege, the only objects that the user has access to in the recycle bin are those that the user owns. A user can view his objects in the recycle bin using the following statement:

SELECT * FROM RECYCLEBIN;

When you drop a tablespace including its contents, the objects in the tablespace are not placed in the recycle bin and the database purges any entries in the recycle bin for objects located in the tablespace. The database also purges any recycle bin entries for objects in a tablespace when you drop the tablespace, not including contents, and the tablespace is otherwise empty. Likewise:

■ When you drop a user, any objects belonging to the user are not placed in the recycle bin and any objects in the recycle bin are purged.

■ When you drop a cluster, its member tables are not placed in the recycle bin and any former member tables in the recycle bin are purged.

■ When you drop a type, any dependent objects such as subtypes are not placed in the recycle bin and any former dependent objects in the recycle bin are purged.

Object Naming the Recycle Bin:

When a dropped table is moved to the recycle bin, the table and its associated objects are given system-generated names. This is necessary to avoid name conflicts that may arise if multiple tables have the same name. This could occur under the following circumstances:

- A user drops a table, re-creates it with the same name, then drops it again.
- Two users have tables with the same name, and both users drop their tables.

The renaming convention is as follows:

BIN$unique_id$version

where:

- unique_id is a 26-character globally unique identifier for this object, which makes the recycle bin name unique across all databases
- version is a version number assigned by the database

Recycle Bin Tables

Enable or Disable Recycle Bin

When the recycle bin is enabled, dropped tables and their dependent objects are placed in the recycle bin. When the recycle bin is disabled, dropped tables and their dependent objects are not placed in the recycle bin; they are just dropped, and you must use other means to recover them (such as recovering from backup).

Disabling the recycle bin does not purge or otherwise affect objects already in the recycle bin. The recycle bin is enabled by default.

You enable and disable the recycle bin by changing the recyclebin initialization parameter. This parameter is not dynamic, so a database restart is required when you change it with an ALTER SYSTEM statement.

Purging Objects in the Recycle Bin

If you decide that you are never going to restore an item from the recycle bin, you can use the PURGE statement to remove the items and their associated objects from the recycle bin and release their storage space. You need the same privileges as if you were dropping the item.

When you use the PURGE statement to purge a table, you can use the name that the table is known by in the recycle bin or the original name of the table. The following hypothetical example purges the table hr.int_admin_emp, which was renamed to

BIN$jsleilx392mk2=293$0 when it was placed in the recycle bin:

PURGE TABLE "BIN$jsleilx392mk2=293$0";

You can achieve the same result with the following statement:

PURGE TABLE int_admin_emp;

You can use the PURGE statement to purge all the objects in the recycle bin that are from a specified tablespace or only the tablespace objects belonging to a specified user, as shown in the following examples:

PURGE TABLESPACE example;

PURGE TABLESPACE example USER oe;

Users can purge the recycle bin of their own objects, and release space for objects, by using the following statement:

PURGE RECYCLEBIN;

If you have the SYSDBA privilege, then you can purge the entire recycle bin by specifying DBA_RECYCLEBIN, instead of RECYCLEBIN in the previous statement.

You can also use the PURGE statement to purge an index from the recycle bin or to purge from the recycle bin all objects in a specified tablespace.

Restoring Objects From Recycle Bin

When you restore a table from the recycle bin, dependent objects such as indexes do not get their original names back; they retain their system-generated recycle bin names. You must manually rename dependent objects to restore their original names. If you plan to manually restore original names for dependent objects, ensure that you make note of each dependent object's system-generated recycle bin name before you restore the table.

The following example restores int_admin_emp table and assigns to it a new name:

FLASHBACK TABLE int_admin_emp TO BEFORE DROP RENAME TO int2_admin_emp;

CS409 MCQS

Topic 105

Question # 1 of 3 **Time Left: 75 sec(s)**

When we want to drop a table from the database, which clause should we use with the drop cluster statement?

Select the correct option

Reload Math Equations

- ( ) Remove Tables
- (•) Including Tables
- ( ) Excluding Tables
- ( ) Update Tables

Click to Save Answer & Move to Next Question

Question # 1 of 3 **Time Left: 87 sec(s)**

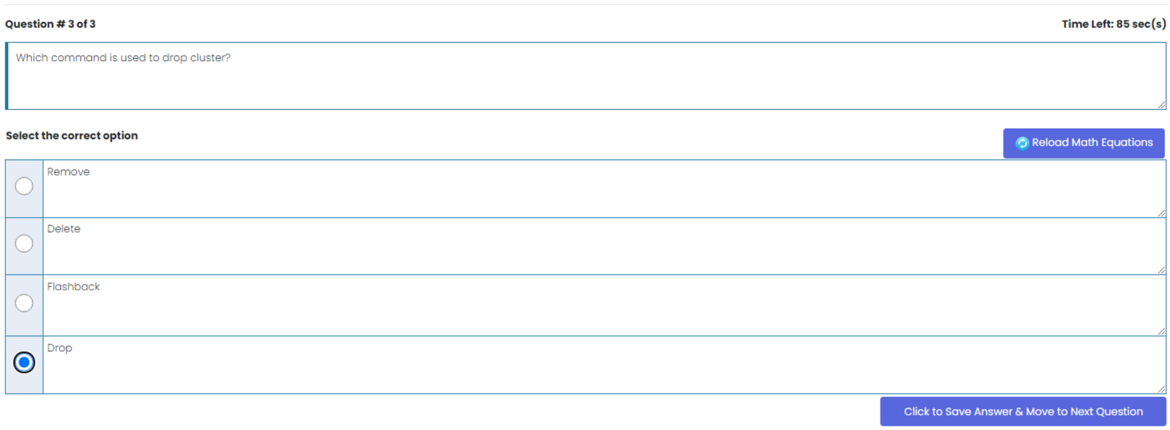
You can alter an existing cluster to change which of the following settings?

Select the correct option

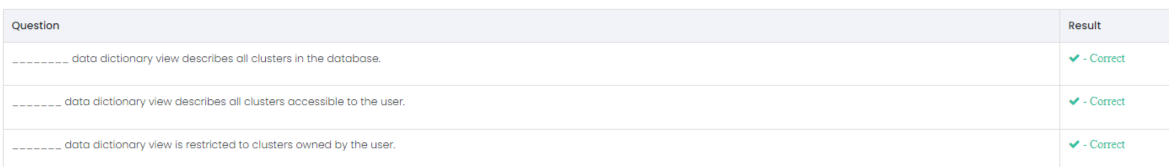
Reload Math Equations

- ( ) None of these
- (•) Physical attributes (INITRANS and storage characteristics)
- ( ) There is no default degree of parallelism
- ( ) The average amount of space is not required to store all the rows for a cluster key value (SIZE)

Click to Save Answer & Move to Next Question



Lesson 106 : Data Dictionary Views for Clusters



Question	Result
________ data dictionary view describes all clusters in the database.	✔ - Correct
_______ data dictionary view describes all clusters accessible to the user.	✔ - Correct
_______ data dictionary view is restricted to clusters owned by the user.	✔ - Correct

DBA

ALL

USER

Lesson 107 : Defining Hash Clusters

What is a Hash Cluster?

Storing a table in a hash cluster is an optional way to improve the performance of data retrieval. A hash cluster provides an alternative to a non-clustered table with an index or an index cluster. With an indexed table or index cluster, Oracle Database locates the rows in a table using key values that the database stores in a separate index. To use hashing, you create a hash cluster and load tables into it. The database physically stores the rows of a table in a hash cluster and retrieves them according to the results of a hash function.

The database physically stores the rows of a table in a hash cluster and retrieves them according to the results of a __________.

Select the correct option

Reload Math Equations

- User-define Function
- Hash Function
- Built-in Function
- None of these

Click to Save Answer & Move to Next Question

Question # 2 of 3 **Time Left: 76 sec(s)**

Storing a table in a hash cluster is an optional way to improve the _______ of data retrieval.

Select the correct option

Reload Math Equations

- Output
- Performance
- Achievement
- Behavior

Click to Save Answer & Move to Next Question

Question # 3 of 3 **Time Left: 86 sec(s)**

Identify the condition where hashing is useful.

Select the correct option

Reload Math Equations

- You cannot afford to preallocate the space that the hash cluster will eventually need.
- Most queries are equality queries on the cluster key
- Most queries on the table retrieve rows over a range of cluster key values.
- The table is not static, but instead is continually growing.

Click to Save Answer & Move to Next Question

Lesson 108 : Create & Alter Hash Clusters

Creating Hash Clusters

You create a hash cluster using a CREATE CLUSTER statement, but you specify a HASHKEYS clause

Identify the correct statement of altering the cluster of employee department.

Select the correct option

Reload Math Equations

- ALTER emp_dep CLUSTER;
- (selected) ALTER CLUSTER emp_dep;
- None of these
- Emp_dep ALTER CLUSTER;

Click to Save Answer & Move to Next Question

Question # 2 of 3 Time Left: 80 sec(s)

You can drop a hash cluster using the ________ statement.

Select the correct option

Reload Math Equations

- Alter Cluster
- Delete Cluster
- (selected) Drop Cluster
- Remove Cluster

Click to Save Answer & Move to Next Question

Question # 3 of 3 Time Left: 86 sec(s)

You create a hash cluster using a CREATE CLUSTER statement, but you have to specify a _______ clause while creating the cluster.

Select the correct option

Reload Math Equations

- TABLESPACE
- HASH
- HASHFUNCTION
- (selected) HASHKEYS

Click to Save Answer & Move to Next Question

Question # 3 of 3 Time Left: 88 sec(s)

______ show information about schema objects that are accessible to you at different level of privilege.

Select the correct option

Reload Math Equations

- Hash Function
- Procedures
- (selected) Data Dictionary Views
- Tablespaces

Click to Save Answer & Move to Next Question

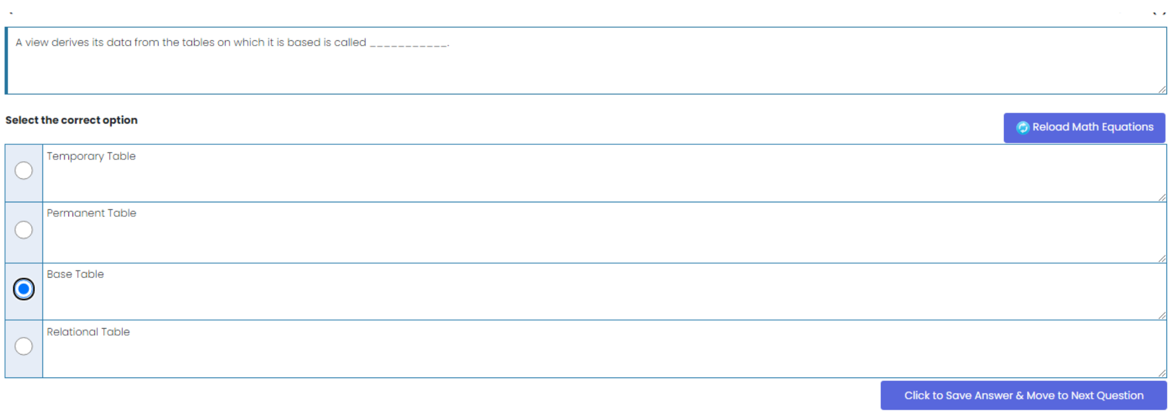
A _____ is a logical representation of a table or combination of tables.

Select the correct option

Reload Math Equations

- ( ) Tablespace
- ( ) Procedures
- (•) View
- ( ) Sequence

Click to Save Answer & Move to Next Question



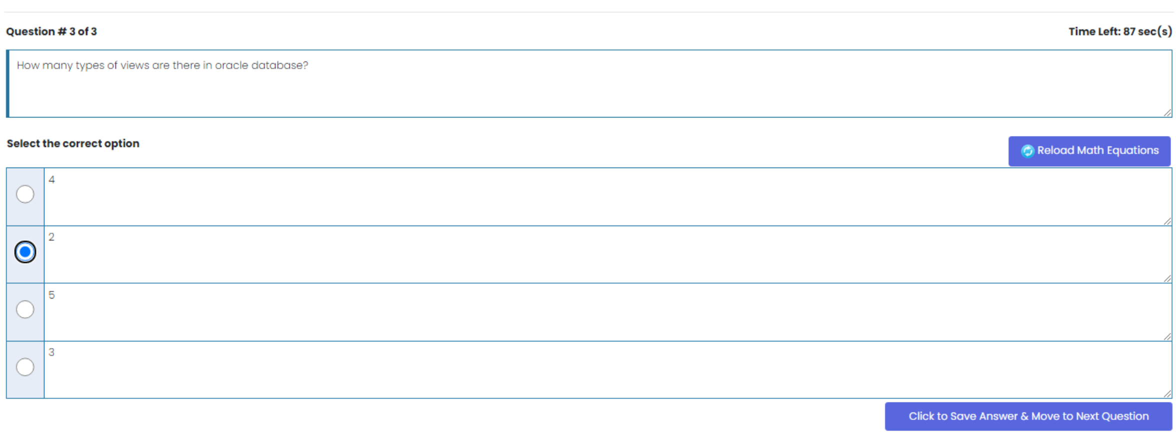
A view derives its data from the tables on which it is based is called ___________.

Select the correct option

Reload Math Equations

- ( ) Temporary Table
- ( ) Permanent Table
- (•) Base Table
- ( ) Relational Table

Click to Save Answer & Move to Next Question



Question # 3 of 3 **Time Left: 87 sec(s)**

How many types of views are there in oracle database?

Select the correct option

Reload Math Equations

- ( ) 4
- (•) 2
- ( ) 5
- ( ) 3

Click to Save Answer & Move to Next Question

Lesson 111 : Updating a Join Views

Updating a Join View

An updatable join view (also referred to as a modifiable join view) is a view that contains multiple tables in the top-level FROM clause of the SELECT statement, and is not restricted by the WITH READ ONLY clause.

Rules for Updating Join Views

The rules for updatable join views are shown in the following table. Views that meet these criteria are said to be inherently updatable.

Rule	Description
General Rule	Any `INSERT`, `UPDATE`, or `DELETE` operation on a join view can modify only one underlying base table at a time.
`UPDATE` Rule	All updatable columns of a join view must map to columns of a **key-preserved table**. See "Key-Preserved Tables" on page 24-7 for a discussion of key-preserved tables. If the view is defined with the `WITH CHECK OPTION` clause, then all join columns and all columns of repeated tables are not updatable.
`DELETE` Rule	Rows from a join view can be deleted as long as there is exactly one key-preserved table in the join. The key preserved table can be repeated in the `FROM` clause. If the view is defined with the `WITH CHECK OPTION` clause and the key preserved table is repeated, then the rows cannot be deleted from the view.
`INSERT` Rule	An `INSERT` statement must not explicitly or implicitly refer to the columns of a **non-key-preserved table**. If the join view is defined with the `WITH CHECK OPTION` clause, `INSERT` statements are not permitted.

Question # 1 of 3 Time Left: 88 sec(s)

How many rules are there in oracle for updating a join view?

Select the correct option Reload Math Equations

- 4
- 3
- 5
- 2

Click to Save Answer & Move to Next Question

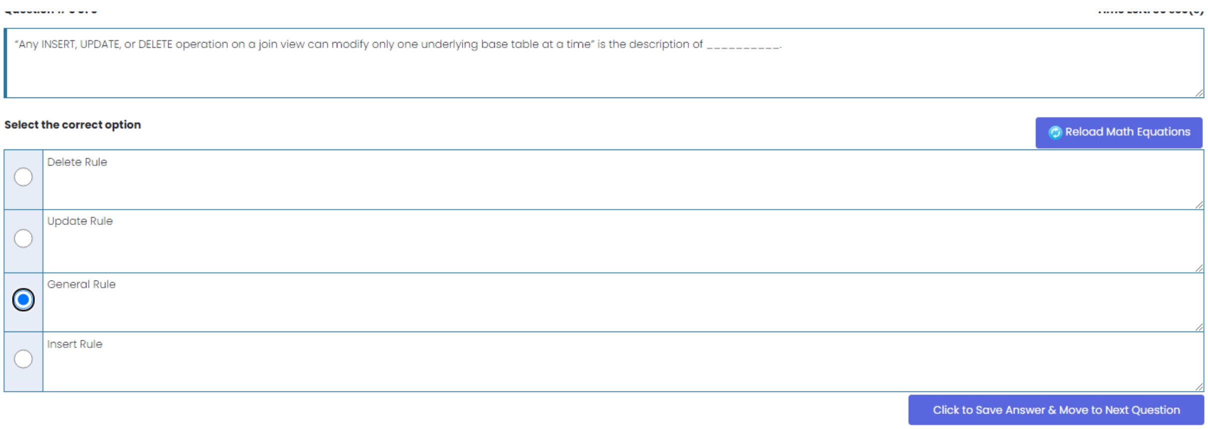
Question # 2 of 3 Time Left: 87 sec(s)

Which of the following is not a rule name for updating a join view in oracle?

Select the correct option Reload Math Equations

- Insert
- Update
- Remove
- Delete

Click to Save Answer & Move to Next Question



Lesson 112 : Create Sequences

Sequences

Sequences are database objects from which multiple users can generate unique integers. The sequence generator generates sequential numbers, which can be used to generate unique primary keys automatically, and to coordinate keys across multiple rows or tables. A sequence generator provides a highly scalable and well-performing method to generate surrogate keys for a number data type.

Sequence Characteristics

A sequence definition indicates general information, such as the following:

- The name of the sequence
- Whether the sequence ascends or descends The interval between numbers
- Whether the database should cache sets of generated sequence numbers in memory
- Whether the sequence should cycle when a limit is reached
- **Data Dictionary Views for Sequence**
- USER_SEQUENCES
- ALL_SEQUENCES
- SEQ

NEXTVAL & CURRVAL

Question # 1 of 3 Time Left: 75 sec(s)

Which of the following option is not available while creating sequence?

Select the correct option

Reload Math Equations

- Maxvalue
- End
- Minvalue
- Start

Click to Save Answer & Move to Next Question

Question # 2 of 3 Time Left: 87 sec(s)

______ are database objects from which multiple users can generate unique integers.

Select the correct option

Reload Math Equations

- Synonyms
- Sequences
- Data Dictionary
- Function

Click to Save Answer & Move to Next Question

Question # 1 of 3 Time Left: 88 sec(s)

Which column is used to get the next value of the sequence?

Select the correct option

Reload Math Equations

- NXTVL
- NEXTVALUE
- NEXT
- NEXTVAL

Click to Save Answer & Move to Next Question

Lesson 113 : Alter & Drop sequences

NEXTVAL & CURRVAL

CURRVAL and NEXTVAL cannot be used in the following places:

- A subquery
- A view query or materialized view query
- A SELECT statement with the DISTINCT operator
- A SELECT statement with a GROUP BY or ORDER BY clause
- A SELECT statement that is combined with another SELECT statement with the UNION, INTERSECT, or MINUS set operator
- The WHERE clause of a SELECT statement
- The condition of a CHECK constraint

Question # 1 of 3 **Time Left: 87 sec(s)**

To drop a sequence in another schema, you must have the ________ system privilege.

Select the correct option Reload Math Equations

- ○ DROP SEQUENCE
- ○ DROP ANY CLUSTER
- ◉ DROP ANY SEQUENCE
- ○ DROP CLUSTER

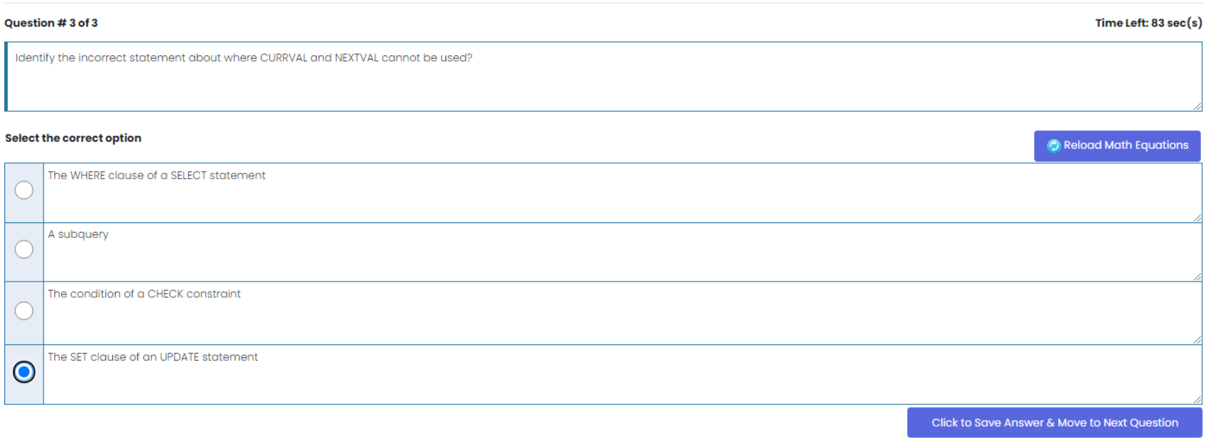
Click to Save Answer & Move to Next Question

Identify the correct statement about where CURRVAL and NEXTVAL cannot be used?

Select the correct option Reload Math Equations

- ○ The SET clause of an UPDATE statement
- ○ VALUES clause of INSERT statements
- ◉ A SELECT statement with a GROUP BY or ORDER BY clause
- ○ The SELECT list of a SELECT statement

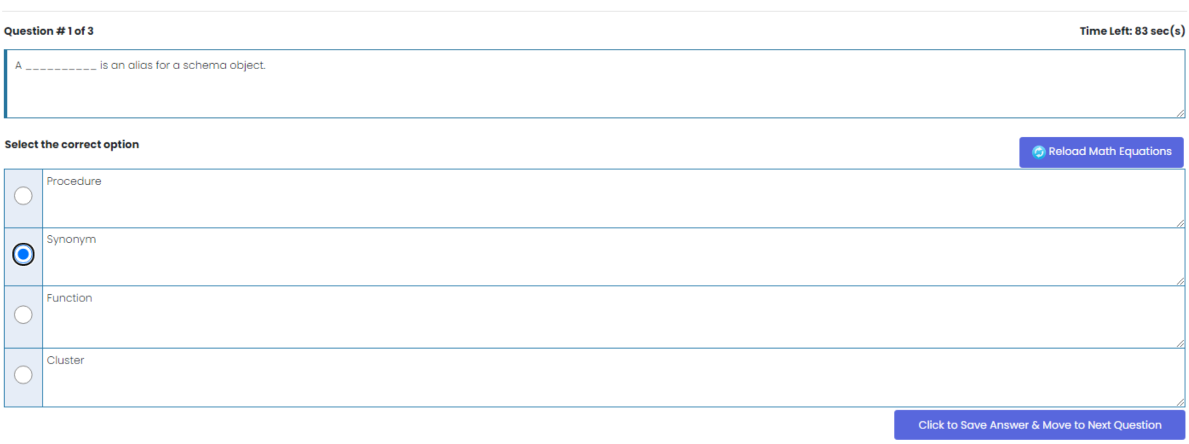
Click to Save Answer & Move to Next Question



Lesson 114 : Defining Synonyms

Public & Private Synonyms

You can create both public and private synonyms. A public synonym is owned by the special user group named PUBLIC and is accessible to every user in a database. A private synonym is contained in the schema of a specific user and available only to the user and to grantees for the underlying object.



Question # 2 of 3 Time Left: 83 sec(s)

Which DDL command is required to create public synonyms?

Select the correct option

Reload Math Equations

- (selected) CREATE PUBLIC SYNONYM
- DROP SYNONYM
- CREATE SYNONYM
- DROP PUBLIC SYNONYM

Click to Save Answer & Move to Next Question

Question # 3 of 3 Time Left: 88 sec(s)

How many types of synonyms are there in oracle database?

Select the correct option

Reload Math Equations

- (selected) 2
- 4
- 3
- 1

Click to Save Answer & Move to Next Question

Question # 1 of 3 Time Left: 86 sec(s)

Which of the following command is used to drop the private synonyms?

Select the correct option

Reload Math Equations

- DROP PUBLIC SYNONYM
- DELETE PRIVATE SYNONYM
- DROP PRIVATE SYNONYM
- (selected) DROP SYNONYM

Click to Save Answer & Move to Next Question

Question # 2 of 3 Time Left: 81 sec(s)

You cannot specify a schema for the synonym if you have specified_________ in command.

Select the correct option

Reload Math Equations

- [ ] None of these
- [ ] Protected
- [ ] Private
- [x] Public

Click to Save Answer & Move to Next Question

Question # 3 of 3 Time Left: 86 sec(s)

Which of the following command is used to drop the public synonyms?

Select the correct option

Reload Math Equations

- [ ] DELETE SYNONYM
- [ ] DELETE PUBLIC SYNONYM
- [x] DROP PUBLIC SYNONYM
- [ ] DROP SYNONYM

Click to Save Answer & Move to Next Question

Question # 1 of 3 Time Left: 79 sec(s)

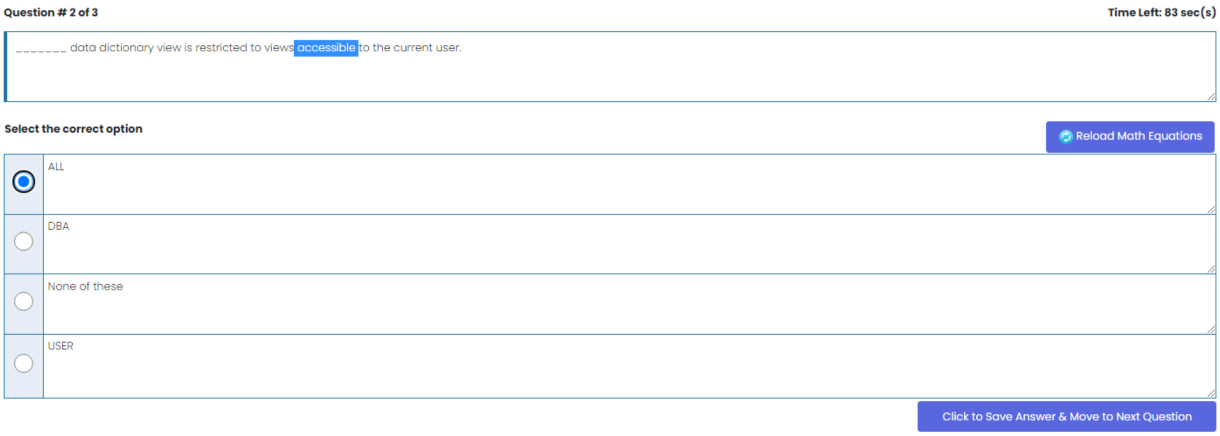
________ data dictionary view describes all views in the database.

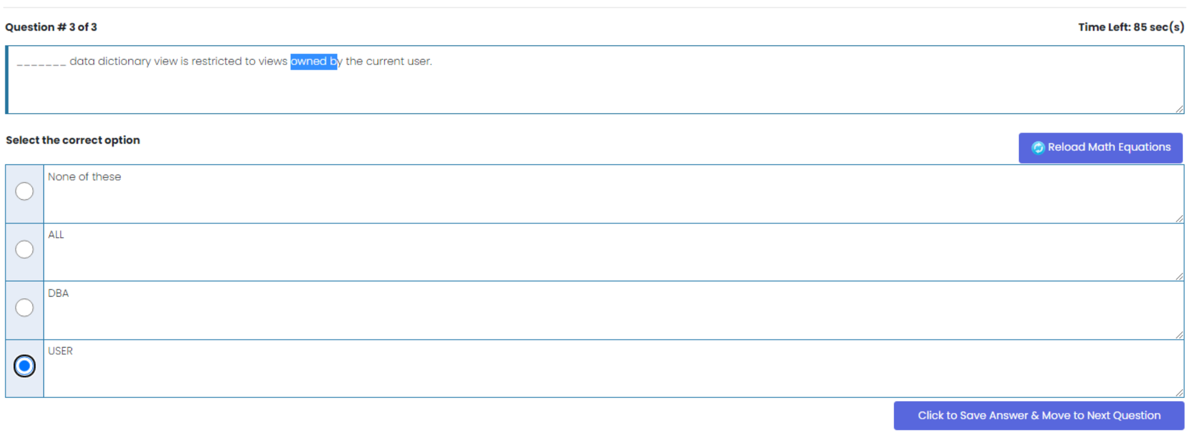
Select the correct option

Reload Math Equations

- [ ] USER
- [ ] None of these
- [ ] ALL
- [x] DBA

Click to Save Answer & Move to Next Question





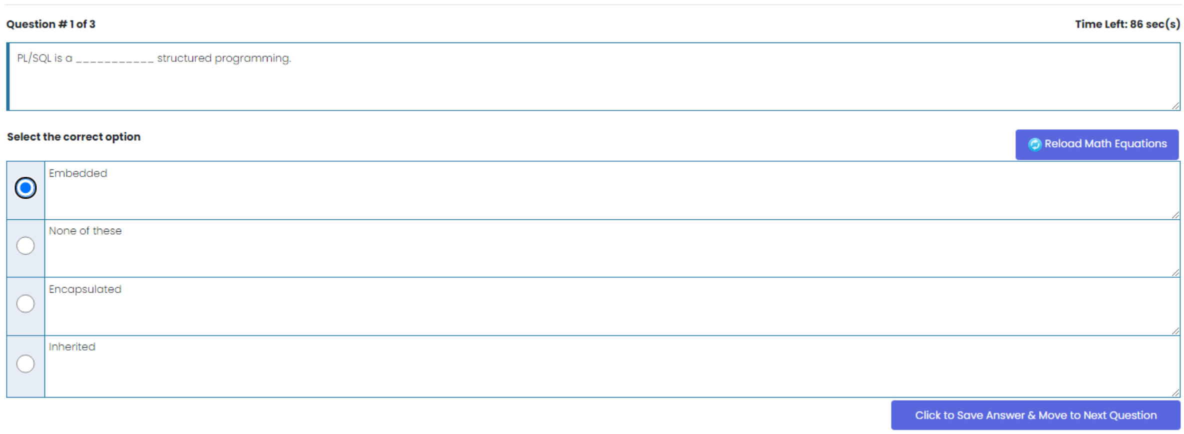
Lesson 117 : Introduction to DB Server Programming using PL/SQL

What is PL/SQL

- Embedded structured programming
- Performed on Server Machine
- Centrally accessible from single database
- Executed on server without any network overhead

PL/SQL & Other Languages

- Embedded SQL (PL/SQL, JAVA/ VB & DB)
- Database Server Level Programming(PL/SQL, Transact-SQL, IBM DB2-Cobol, ProC, ProCobol)
- (Front end programming can be changed without affecting Server programming)
- Database Client Programming (Developer 9i, JDeveloper 9i, Java (J2EE), VB, .Net)



Question # 1 of 3

Time Left: 86 sec(s)

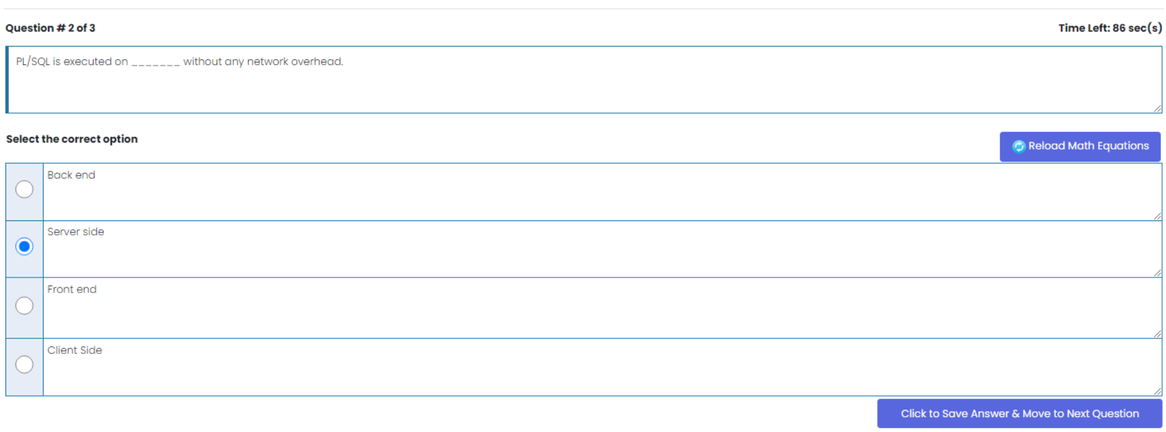
PL/SQL is a ___________ structured programming.

Select the correct option

Reload Math Equations

- (selected) Embedded
- None of these
- Encapsulated
- Inherited

Click to Save Answer & Move to Next Question



Question # 2 of 3

Time Left: 86 sec(s)

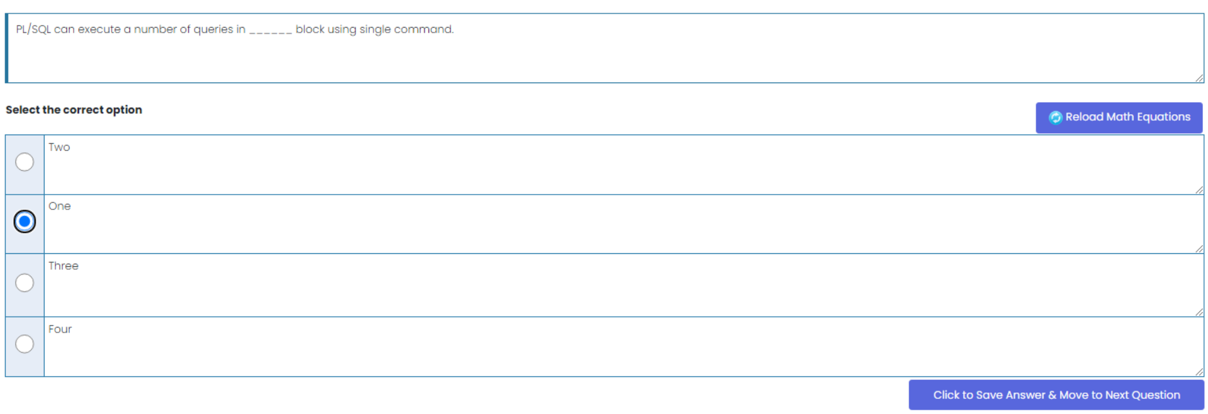
PL/SQL is executed on _______ without any network overhead.

Select the correct option

Reload Math Equations

- Back end
- (selected) Server side
- Front end
- Client Side

Click to Save Answer & Move to Next Question



PL/SQL can execute a number of queries in ______ block using single command.

Select the correct option

Reload Math Equations

- Two
- (selected) One
- Three
- Four

Click to Save Answer & Move to Next Question

Lesson 119 : PL/SQL Benefits & Basic Constructs

PL/SQL Benefits

More powerful than pure SQL because it combines the power of SQL and

- Iteration (loops)

    - A loop statement allows us to execute a statement or group of statements multiple times.
- Selection (Ifs)
- Cursors
    - A cursor is a pointer to this context area. PL/SQL controls the context area through a cursor. A cursor holds the rows (one or more) returned by a SQL statement. The set of rows the cursor holds is referred to as the active set.
- Block Structures
    - PL/SQL blocks have a pre-defined structure in which the code is to be grouped. Below are different sections of PL/SQL blocks.

Three Sections:-

Declaration Section (DELCLARE)

Execution Section (BEGIN,END)

Exception Handling (EXCEPTION ,END)

PL/SQL Basic Constructs

Basic Structure

Running a program

Variables

SELECT INTO

Comments

IFs

LOOPs

Output

Question # 1 of 3 Time Left: 87 sec(s)

PL/SQL is more powerful than pure__________.

Select the correct option

Reload Math Equations

- C#
- Python
- Database
- SQL

Click to Save Answer & Move to Next Question

Question # 2 of 3 Time Left: 87 sec(s)

A __________ statement allows us to execute a statement or group of statements multiple times.

Select the correct option

Reload Math Equations

- Loop
- If else
- Selection
- None of these

Click to Save Answer & Move to Next Question

Question # 3 of 3 Time Left: 88 sec(s)

A cursor is a _______ to this context area.

Select the correct option

Reload Math Equations

- Pointer
- Function
- Variable
- Integer

Click to Save Answer & Move to Next Question

Question # 1 of 3

Time Left: 82 sec(s)

In basic structure of PL/SQL execution section starts with _____ keyword.

Select the correct option

Reload Math Equations

- END
- EXCEPTIONS
- BEGIN (selected)
- DECLARE

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 88 sec(s)

In basic structure of PL/SQL execution section end with _____ keyword.

Select the correct option

Reload Math Equations

- END (selected)
- DECLARE
- BEGIN
- EXCEPTIONS

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 87 sec(s)

In basic structure of PL/SQL exception section starts with ______ keyword.

Select the correct option

Reload Math Equations

- DECLARE
- EXCEPTIONS (selected)
- END
- BEGIN

Click to Save Answer & Move to Next Question

Question # 1 of 3 **Time Left: 74 sec(s)**

Which of the following is correct syntax to print the output in PL/SQL?

Select the correct option

Reload Math Equations

- db_output.put_line(t.table_name);
- dbms_output_line(t.table_name);
- dbms_output.put_line(t.table_name);
- dbms_output.put_(t.table_name);

Click to Save Answer & Move to Next Question

Question # 2 of 3 **Time Left: 74 sec(s)**

Which of the following is correct syntax for assigning value 10 to a variable X?

Select the correct option

Reload Math Equations

- X:=10
- X=:10
- X=10
- X==10

Click to Save Answer & Move to Next Question

Question # 3 of 3 **Time Left: 85 sec(s)**

Which of the following is correct output for the following PL/SQL program?

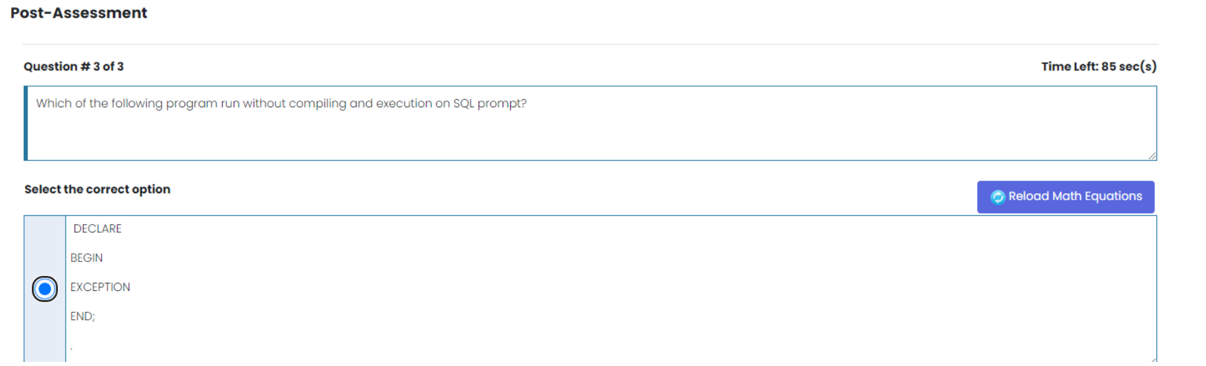
```
DECLARE
   msg varchar2(40) := 'I love Oracle Programing' ;
BEGIN
   dbms_output.put_line(msg);
END;
/
```

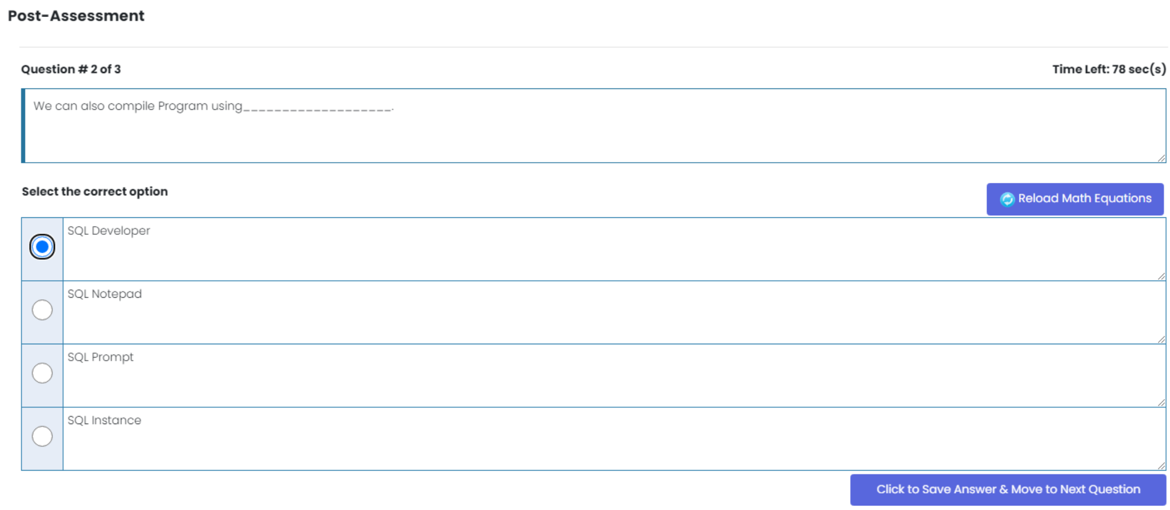
Select the correct option

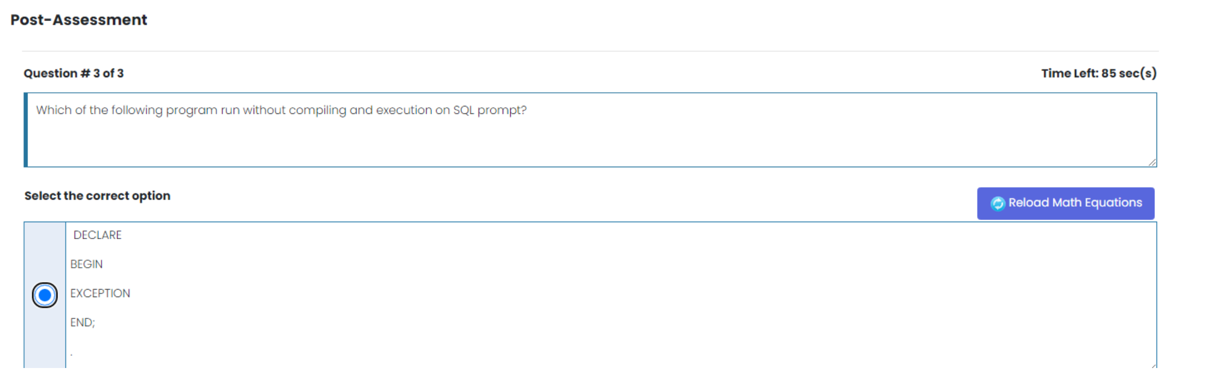
Reload Math Equations

- Msg
- I love Oracle Programing
- (msg)
- 'I love Oracle Programing';

Click to Save Answer & Move to Next Question







Lesson 124 : Data Types & Select INTO Clause

Introduction

- Each value in PL/SQL such as a constant, variable and parameter has a data type that determines the storage format, valid values, and allowed operations.
- PL/SQL has two kinds of data types: scalar and composite.

Scalar Data Types Families

- Number
- Boolean
- Character
- Datetime

A scalar data type may have subtypes. A subtype is a data type that is a subset of another data type, which is its base type. A subtype further defines a base type by restricting the value or size of the base data type.

Numeric Data Types

- The numeric data types represent real numbers, integers, and floating-point numbers. They are stored as NUMBER, IEEE floating-point storage types (BINARY_FLOAT and BINARY_DOUBLE), and PLS_INTEGER.
- The data types NUMBER, BINARY_FLOAT, and BINARY_DOUBLE are SQL data types.
- The PLS_INTEGER datatype is specific to PL/SQL. It represents signed 32 bits int.

Boolean Data Types

- The BOOLEAN datatype has three data values: TRUE, FALSE, and NULL. Boolean values are typically used in control flow structure such as IF-THEN, CASE, and loop statements like LOOP, FOR LOOP, and WHILE LOOP.
- SQL does not have the BOOLEAN data type, you can assign data type NUMBER(1) or CHAR(1)

Character or String Data Types

- The character data types represent alphanumeric text. PL/SQL uses the SQL character data types such as
- CHAR, VARCHAR2, LONG, RAW, LONG RAW, ROWID, and UROWID
- CHAR is used for fixed length columns such primary keys or Boolean.

DateTime Data Types

- The datetime data types represent dates, timestamp with or without time zone and intervals. PL/SQL datetime data types are DATE, TIMESTAMP, TIMESTAMP WITH TIME ZONE, TIMESTAMP WITH LOCAL TIME ZONE, INTERVAL YEAR TO MONTH, and INTERVAL DAY TO SECOND

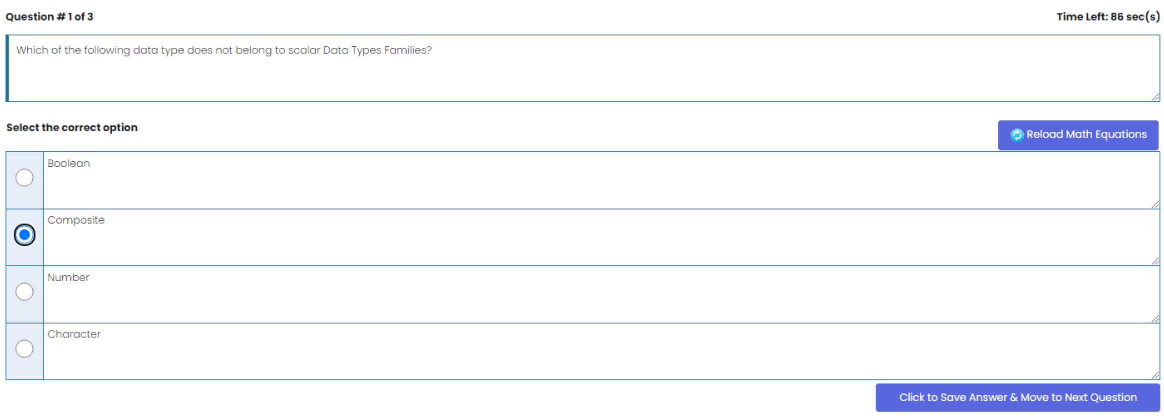
Composite Data Types

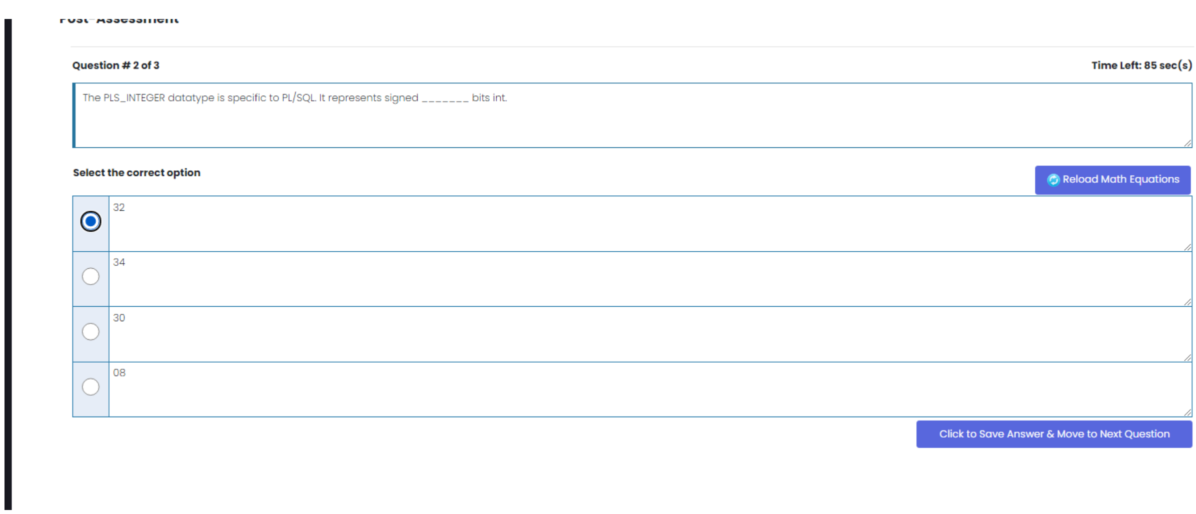
- TableName.Column%TYPE is a composite data type. For example, EMP.sal%TYPE
- It will follow same data type and size of column of a table.
- In case if size or type of column is changed then there is no need to re-compile the program

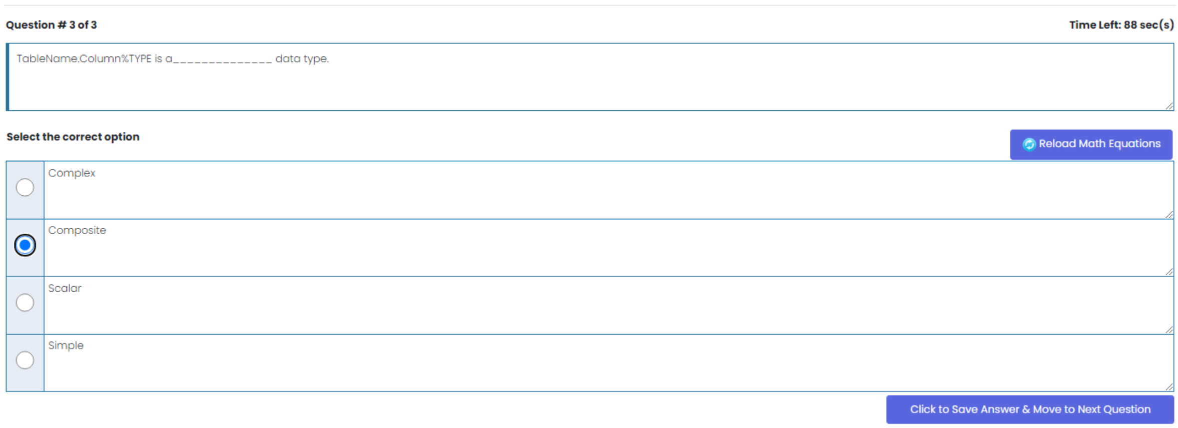
Data Types Synonyms

SELECT INTO Clause

We cannot use simple SQL Select clause statement, rather we use Select INTO by storing values in variables with condition of choosing single record







Lesson 126 : Control IF Conditions

IF THEN Condition

There are three ways to write IF condition:

- IF – END IF
- IF – ELSE – END IF
- IF – ELSIF – ELSE – END IF

Structure of IF THEN Condition

There are three ways to write IF condition:

IF condition **THEN**

statements;

END IF;

For example:

IF v_sal > 2000 Then

total_sal := v_sal + 5000;

END IF;

Example of IF THEN Condition

Structure of IF THEN ELSE Condition

There are three ways to write IF condition:

IF condition **THEN**

statements;

ELSE

statements;

END IF;

Example of IF THEN ELSE Condition

Structure of IF THEN ELSIF Condition

There are three ways to write IF condition:

IF condition **THEN**

statements;

ELSIF

statements;

ELSE

statements;

END IF;

Example of IF THEN ELSIF Condition

LOOPS

- LOOP – EXIT WHEN – END LOOP
- FOR – LOOP – END LOOP
- WHILE – LOOP – END LOOP

Question # 1 of 3 **Time Left: 78 sec(s)**

Which of the following is correct Syntax of For LOOP?

Select the correct option Reload Math Equations

- ( ) FOR index IN lower_bound .. upper_bound LOOP
  statements;
  LOOP;
- (•) FOR index IN lower_bound .. upper_bound LOOP
  statements;
  END LOOP;
- ( ) FOR index IN lower_bound .. upper_bound
  statements;
  END LOOP;
- ( ) FOR index lower_bound .. upper_bound LOOP
  statements;
  END LOOP;

Click to Save Answer & Move to Next Question

Question # 2 of 3 **Time Left: 88 sec(s)**

A ______ statement allows us to execute a statement or group of statements multiple times.

Select the correct option Reload Math Equations

- (•) Loop
- ( ) Condition
- ( ) Array
- ( ) Procedure

Click to Save Answer & Move to Next Question

Question # 3 of 3 **Time Left: 87 sec(s)**

Which of the following is not a syntax for while loop in PL/SQL?

Select the correct option Reload Math Equations

- (•) WHILE –END LOOP
- ( ) FOR – LOOP – END LOOP
- ( ) WHILE – LOOP – END LOOP
- ( ) LOOP – EXIT WHEN – END LOOP

Click to Save Answer & Move to Next Question

Lesson 129 : Exception Handlings-1

PL/SQL Exception

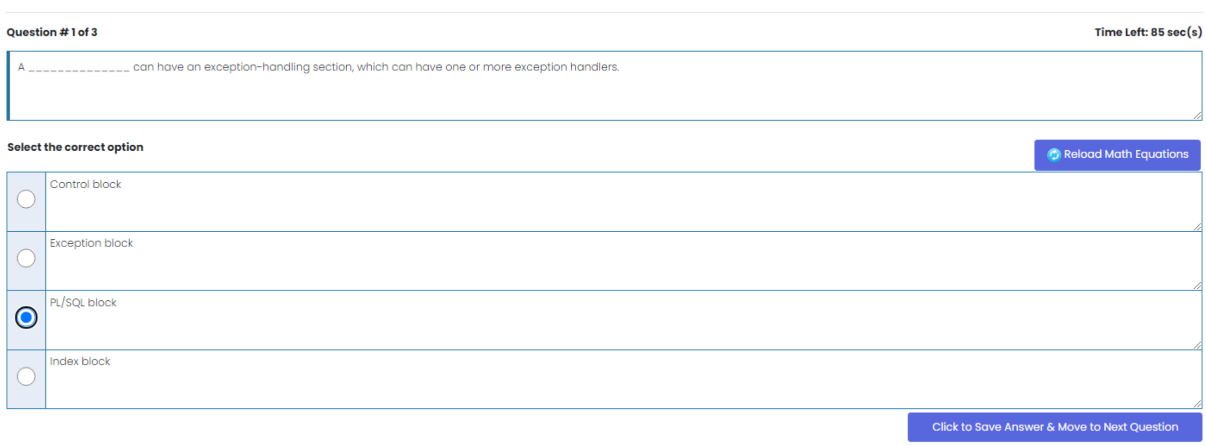
- PL/SQL treats all errors that occur in an anonymous block, procedure, or function as exceptions.
- The exceptions can have different causes such as coding mistakes, bugs, even hardware failures.
- It is not possible to anticipate all potential exceptions, however, you can write code to handle exceptions to enable the program to continue running as normal.

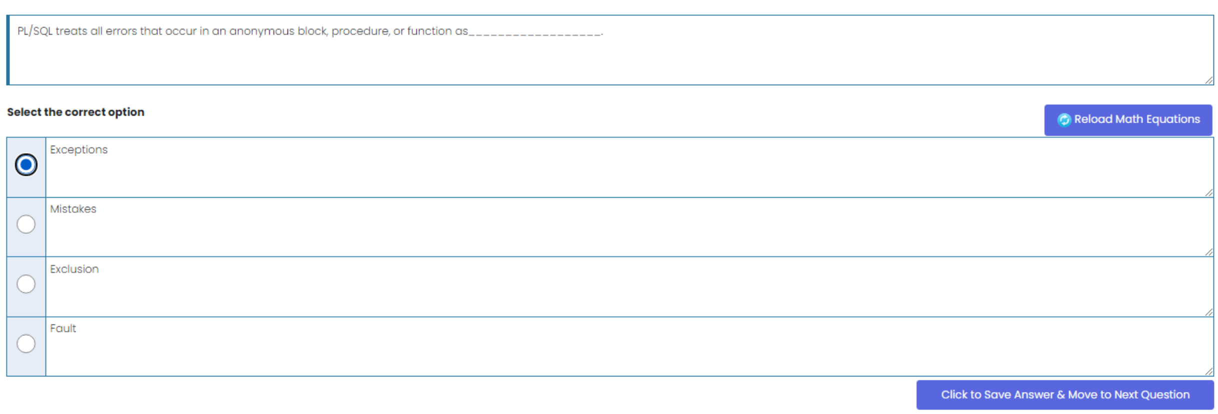
PL/SQL Exception

- The code that you write to handle exceptions is called an exception handler.
- A PL/SQL block can have an exception-handling section, which can have one or more exception handlers.

Raise Exception

Exceptions are raised by the database server automatically whenever there is any internal database error, but exceptions can be raised explicitly by the programmer by using the command **RAISE**. Following is the simple syntax for raising an exception:





What is a Procedure in PL/SQL?

- Is a block with a name
- The DECLARE key word is not used
- Parameters can be

IN

OUT

IN OUT

Is stored (USER_SOURCE)

Question # 2 of 3

Time Left: 87 sec(s)

___________is a block with a name in PL/SQL.

Select the correct option

Reload Math Equations

- (selected) Procedure
- Index
- Control file
- Exception

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 87 sec(s)

In PL/SQL procedure, _____________ keyword is not used.

Select the correct option

Reload Math Equations

- DELETE
- (selected) DECLARE
- ADD
- END

Click to Save Answer & Move to Next Question

Question # 1 of 3

Time Left: 83 sec(s)

Which of the following is only parameter for the PL/SQL Function?

Select the correct option

Reload Math Equations

- IN OUT
- OUT
- OUT IN
- (selected) IN

Click to Save Answer & Move to Next Question

Question # 2 of 3 **Time Left: 81 sec(s)**

__________ returns one value only.

Select the correct option

Reload Math Equations

- (selected) PL/SQL Function
- PL/SQL Control block
- PL/SQL Block
- PL/SQL Procedure

Click to Save Answer & Move to Next Question

Question # 3 of 3 **Time Left: 84 sec(s)**

In PL/SQL function, ____________ keyword is not used.

Select the correct option

Reload Math Equations

- DELETE
- (selected) DECLARE
- END
- ADD

Click to Save Answer & Move to Next Question

Question # 1 of 2 **Time Left: 87 sec(s)**

________________can be called after compilation in some other function, procedure, block or package by assigning it to variable.

Select the correct option

Reload Math Equations

- Procedure
- Index
- (selected) Function
- Table

Click to Save Answer & Move to Next Question

ıssessment

tion # 2 of 2 Time Left: 89 sec(s)

_________can be called after compilation in some other function, procedure, block or package.

t the correct option

Reload Math Equations

- Index
- Table
- Procedure
- Function

Click to Save Answer & Move to Next Question

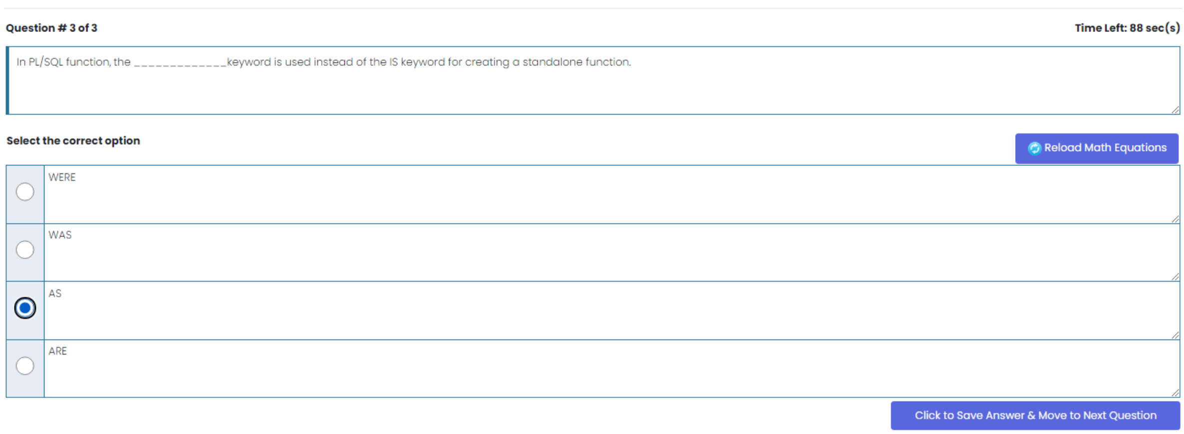
What will be the final output for the following PL/SQL code?

```
CREATE OR REPLACE FUNCTION fact(x number)
RETURN number IS
   f number;
BEGIN
   IF x=0 THEN
      f := 1;
   ELSE
      f := x * fact(x-1);
   END IF;
RETURN f;
END;
/

DECLARE
   num number;
   factorial number;
BEGIN
   num:= 6;
   factorial := fact(num);
   dbms_output.put_line(' Factorial '|| num || ' is ' || factorial);
END;
/
```

Select the correct option

- Factorial num 6 is 720
- 720
- Factorial 6 is 720
- Factorial is 720



Purpose of PL/SQL Package

- In PL/SQL, a package is a schema object that contains definitions for a group of related functionalities.
- A package includes variables, constants, cursors,

exceptions, procedures, functions, and

subprograms.

- It is compiled and stored in the Oracle Database.

PL/SQL Package contains …

- Typically, a package has a specification and a body. A package specification is mandatory while the package body can be required or optional, depending on the package specification.

Schematic Diagram of Package

PL/SQL Package Specification

- The package specification declares the public objects that are accessible from outside the package.
- If a package specification whose public objects include cursors and subprograms, then it must have a body which defines queries for the cursors and code for the subprograms.

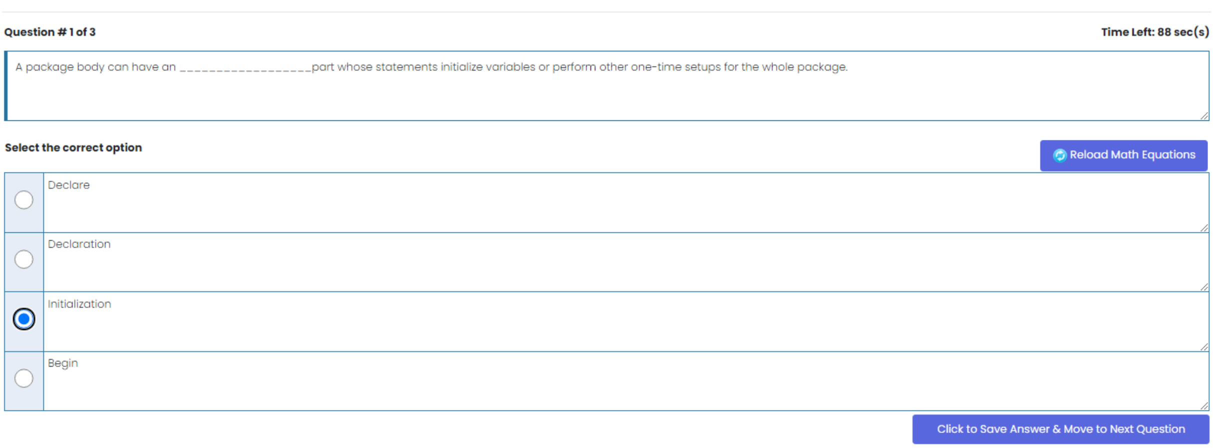
PL/SQL Package Body

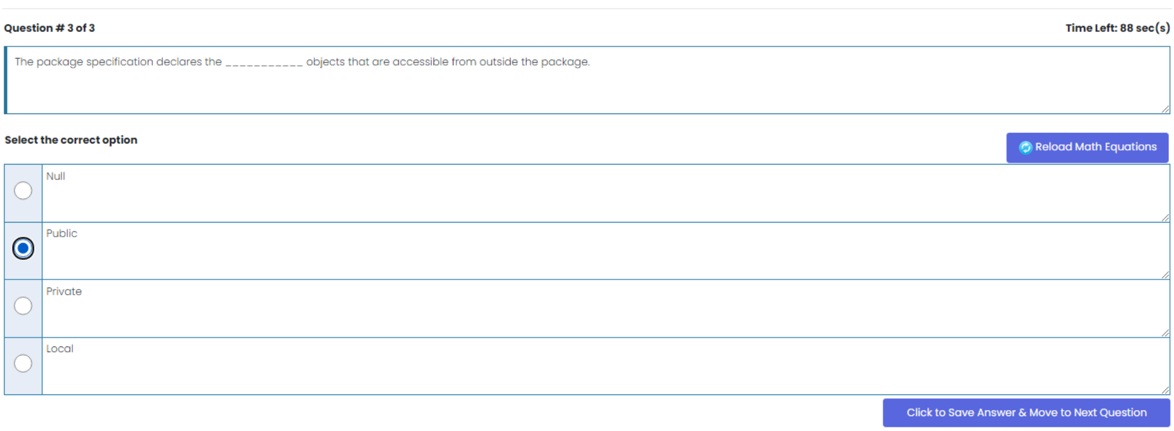
- A package body contains the implementation of the cursors or subprograms declared in the package specification. In the package body, you can declare or define private variables, cursors, etc., used only by package body itself.

PL/SQL Package Body

- A package body can have an initialization part whose statements initialize variables or perform other one-time setups for the whole package.
- A package body can also have an exception-handling part used to handle exceptions.

Package Specification & its Body





Execution of Packages

Compiling a Package using SQL*Plus

Compiling a Package using Oracle Developer

Compiling a Package using Files on SQL*Plus

Calling Function and Procedure from Package

Compiling a Package using SQL*Plus

If you use SQL*Plus for compiling and creating a package body, you type forward slash (/) as follows:

TOPIC 139